

YUSHI HUANG

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EDUCATION

The Hong Kong University of Science and Technology

Ph.D. student at ECE, advised by Prof. [Jun Zhang](#)

2025.02 – Present

Hong Kong SAR, China

Beihang University

Bachelor of Computer Science and Technology

2020.09 – 2024.06

Beijing, China

- Grade Point Average (GPA): 3.86/4.00
- Weighted Score: 93.2/100

RESEARCH INTEREST

My research interest is building efficient and high-performing generative systems. I currently work on RL for efficient image/video generation. Previously, I worked on improving inference efficiency for vision and language generative models, including low-precision inference, computation skipping, efficient attention, etc.

PUBLICATIONS

“*” and “†” denote equal contributions and corresponding authors, respectively.

1. MoDES: Accelerating Mixture-of-Experts Multimodal Large Language Models via Dynamic Expert Skipping 📄 🌐

Yushi Huang, Zining Wang, Zhihang Yuan†, Yifu Ding, Ruihao Gong, Jinyang Guo, Xianglong Liu, Jun Zhang†
Conference on Computer Vision and Pattern Recognition (CVPR), 2026

2. LINVIDEO: A Post-Training Framework towards $\mathcal{O}(n)$ Attention in Efficient Video Generation 📄

Yushi Huang, Xingtong Ge, Ruihao Gong†, Chengtao Lv, Jun Zhang†
Conference on Computer Vision and Pattern Recognition (CVPR), 2026 (**Highlight**)

3. QVGen: Pushing the Limit of Quantized Video Generative Models 📄 🌐

Yushi Huang, Ruihao Gong†, Jing Liu, Yifu Ding, Chengtao Lv, Haotong Qin, Jun Zhang†
International Conference on Learning Representations (ICLR), 2026 (**Ranked in the top 1.3%**)

4. Temporal Feature Matters: A Framework for Diffusion Model Quantization 📄 🌐

Yushi Huang, Ruihao Gong, Xianglong Li†, Jing Liu, Yuhang Li, Jiwen Lu, Dacheng Tao
Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025

5. HarmoniCa: Harmonizing Training and Inference for Better Feature Caching in Diffusion Transformer Acceleration 📄 🌐

Yushi Huang*, Zining Wang*, Ruihao Gong†, Jing Liu, Xinjie Zhang, Jinyang Guo, Xianglong Liu, Jun Zhang†
International Conference on Machine Learning (ICML), 2025

6. TFMQ-DM: Temporal Feature Maintenance Quantization for Diffusion Models 📄 🌐

Yushi Huang*, Ruihao Gong*, Jing Liu, Tianlong Chen, Xianglong Li†
Conference on Computer Vision and Pattern Recognition (CVPR), 2024 (**Highlight**)

7. LLMC: Benchmarking Large Language Model Quantization with a Versatile Compression Toolkit 📄 🌐

Ruihao Gong*, Yang Yong*, Shiqiao Gu*, Yushi Huang*, Chengtao Lv, Yunchen Zhang, Dacheng Tao, Xianglong Li†
Conference on Empirical Methods in Natural Language Processing: Industry Track (EMNLP Industry Track), 2024

8. Flash-VAED: Plug-and-Play VAE Decoders for Efficient Video Generation 📄 🌐

Lunjie Zhu, Yushi Huang, Xingtong Ge, Yufei Xue, Zhening Liu, Yumeng Zhang, Zehong Lin, Jun Zhang†
International Conference on Machine Learning (ICML), 2026

9. Focus-dLLM: Accelerating Long-Context Diffusion LLM Inference via Confidence-Guided Context Focusing 📄 🌐

Lingkun Long, Yushi Huang, Shihao Bai, Ruihao Gong, Jun Zhang, Ao Zhou, Jianlei Yang†
Association for Computational Linguistics (ACL), 2026 (Main)

10. LIGHT FORCING: Accelerating Autoregressive Video Diffusion via Sparse Attention 📄 🌐

Chengtao Lv, Yumeng Shi, Yushi Huang, Ruihao Gong†, Shen Ren, Wenya Wang†
International Conference on Machine Learning (ICML), 2026

11. **LLMC+: Benchmarking Vision-Language Model Compression with a Plug-and-play Toolkit**  
Chengtao Lv, Bilang Zhang, Yang Yong, Ruihao Gong†, **Yushi Huang**, Shiqiao Gu, Jiajun Wu, Yumeng Shi, Jinyang Guo, Wenya Wang†
Association for the Advancement of Artificial Intelligence (AAAI), 2026
12. **SlimInfer: Accelerating Long-Context LLM Inference via Dynamic Token Pruning**  
Lingkun Long, Rubing Yang, **Yushi Huang**, Desheng Hui, Ao Zhou†, Jianlei Yang†
Association for the Advancement of Artificial Intelligence (AAAI), 2026
13. **SGMD: Score Gradient Matching Distillation for Few-Step Video Diffusion Distillation**  
Zhuguanyu Wu, Ruihao Gong†, Yang Yong, **Yushi Huang**, Xiangyu Fan, Lei Yang, Dahua Lin, Xianglong Liu
International Conference on Machine Learning (ICML), 2026
14. **PTSBench: A Comprehensive Post-Training Sparsity Benchmark Towards Algorithms and Models**  
Zining Wang, Jinyang Guo, Yang Yong, Ruihao Gong, Aishan Liu, **Yushi Huang**, Jiaheng Liu, Xianglong Liu†
ACM International Conference on Multimedia (ACM MM), 2024

PREPRINT

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1. **Reinforcing Few-step Generators via Reward-Tilted Distribution Matching**  
Yushi Huang*, Xiangxin Zhou*, Ruoyu Wang*, Chi Zhang, Jun Zhang, Tianyu Pang†
In submission to Conference on Neural Information Processing Systems (NeurIPS), 2026
2. **VLMQ: Token Saliency-Driven Post-Training Quantization for Vision-language Models** 
Yufei Xue, **Yushi Huang**, Jiawei Shao, Lunjie Zhu, Chi Zhang, Xuelong Li†, Jun Zhang†
In submission to European Conference on Computer Vision (ECCV), 2026
3. **ProQuant: Progressive Quantization-aware Training for Edge MLLMs**
Yufei Xue, **Yushi Huang**, Jiawei Shao, Pingcheng Dong, Yonghao Tan, Shiyao Li, Kwang-Ting Cheng, Xuelong Li, Jun Zhang†
In submission to Conference on Neural Information Processing Systems (NeurIPS), 2026
4. **SALT: Self-Consistent Distribution Matching with Cache-Aware Training for Few-Step Video Generation**  
Xingtong Ge, Yi Zhang, **Yushi Huang**, Dailan He, Xiahong Wang, Bingqi Ma, Guanglu Song, Yu Liu, Jun Zhang†
In submission to European Conference on Computer Vision (ECCV), 2026
5. **Exploring the Design Space of Reward Backpropagation for Flow Matching** 
Ruoyu Wang*, Boye Niu*, Xiangxin Zhou*, **Yushi Huang**, Tongliang Liu, Chi Zhang†
In submission to International Conference on Learning Representations (ICLR), 2027
6. **Feed-Forward 3D Gaussian Splatting Compression with Long-Context Modeling** 
Zhening Liu*, Rui Song*, **Yushi Huang**, Yingdong Hu, Xinjie Zhang, Jiawei Shao, Zehong Lin, Jun Zhang†
In submission to European Conference on Computer Vision (ECCV), 2026
7. **Real-Time Human Frontal View Synthesis from a Single Image**  
Fangyu Lin, Yingdong Hu, Lunjie Zhu, Zhening Liu, **Yushi Huang**, Zehong Lin, Jun Zhang†
In submission to European Conference on Computer Vision (ECCV), 2026

PROJECTS

LightCompress: Towards Accurate and Efficient AIGC Model Compression (700+ Stars)

One of the core contributors who:

- Implements many quantization methods for LLM, like QuaRot, GPTQ, SmoothQuant, OmniQuant, etc;
- Builds an end-to-end LLM quantization tool that supports multiple model architectures, evaluation approaches, and inference backends;
- Provides best practices for quantization on LLM under different setups.

EXPERIENCE

Tencent | Hunyuan

Research Intern, mentored by [Tianyu Pang](#)
RL for image/video generative models.

2026.02 – Now
Shenzhen, China

ByteDance | Seed

Research Intern, mentored by Yang Zhang and [Zhihang Yuan](#)
Inference acceleration for multimodal LLMs.

2025.09 – 2025.11
Shenzhen, China

Microsoft Research Asia

Research Intern, mentored by [Fangyun Wei](#)

Video generation and world models.

2024.12 – 2025.02

Beijing, China

SenseTime Research

Research Intern, mentored by [Ruihao Gong](#)

Compression and acceleration for image/video diffusion models and LLMs.

2023.05 – 2025.03

Beijing & Shenzhen, China

ACADEMIC SERVICES

- **Conference Reviewer:** NeurIPS, ICLR, ICML, COLM, AAAI, CVPR, ECCV

SKILLS

- **Programming Languages:** Python, C, Java
- **Scientific Packeags:** Pytorch, Numpy